GGG-N-1907
June 4, 1975
SUPERSEDING
MIL-N-26511C
June 29, 1973

FEDERAL SPECIFICATION

NUTRUNNER AND SCREWDRIVER, PNEUMATIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable, pneumatic nutrunners and screwdrivers.

1.1.1 Federal specification coverage. This specification does not cover all types, styles, and sizes of pneumatic nutrunners and screwdrivers that are commercially available but is intended to cover only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and sizes. Nutrunners and screwdrivers shall be of the following types and sizes as specified (see 6.2).

Type I - Pistol grip handle.

Size 1 - 10 to 50 inch-pounds.
Size 2 - 25 to 90 inch-pounds.

Type II - Body grip handle.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard :

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or directed by the contracting officer.)

FSC 5130

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standard for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

American National Standards Institute (ANSI), Inc., Standard:

B107.4 - Driving and Spindle Ends for Portable Hand Air and Electric Tools.
S5.1 - CAGI-PNEUROP Test Code for the Measurement of Sound from Pneumatic Equipment.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Preproduction model. When specified (see 6.2), the supplier shall furnish within the time frame specified (see 6.2), one nutrunner and screwdriver of the type and style specified in the contract to prove, prior to starting production, that his production methods and choice of design detail will produce tools that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model during production shall be subject to the approval of the contracting officer. Approval of the preproduction model by the contracting agency shall not relieve the supplier of his obligation to furnish tools conforming to this specification.

3.2 Illustrations. The illustrations shown herein are descriptive, not restrictive, and are not intended to preclude the purchase of nutrunners and screwdrivers otherwise conforming to this specification.

3.3 General construction. The nutrunner and screwdriver shall be suitable for use in installing and removing threaded fasteners when equipped with the appropriate driving accessory and operating on an air pressure of 90 PSIG. The motor shall be the rotary vane type equipped with ball bearings. Sleeve bearings are permitted only at the forward end of the clutch housing, and needle bearings are not permitted in thrust applications. A means for applying the lubricant shall be provided and the specific lubricant shall be applied as recommended by the manufacturer.

3.3.1 All working parts subject to wear shall be readily accessible for adjustment or replacement. All parts having the same manufacturer's part number shall be constructed to definite standards in order that they shall be completely interchangeable without modification. All threaded parts shall be in the inch system and conform to the applicable requirements of NBS-H28. All fastening devices such as screws, pins, and similar parts shall be installed with a means to prevent loss of tightness but shall not be permanently deformed, as installed, so as to prevent removal for service or repair. Cadmium is not an acceptable plating on any component.

3.3.2 Materials. The materials used in the construction of all components shall be sound, of uniform quality and condition, and shall be capable of completing the examinations and tests specified herein. All components shall be new and unused. Repaired or reconditioned components are not acceptable.

3.4 Sound pressure level. The sound pressure level shall not exceed 90dBA.

3.5 Handle and throttle. The nutrunner and screwdriver shall have a pistol grip or body grip handle as specified herein. The pistol grip handle shall have a trigger or button type throttle and the body grip handle shall have a lever type throttle. The throttles shall control the volume of air admitted into the motor, automatically shut off the air supply when hand pressure is released and shall be nonleakable (see 4.7.6).

3.6 Reversing switch. The nutrunner and screwdriver shall be equipped with a switch that shall change the direction of rotation of the spindle. The switch shall be located convenient to the operator and shall maintain one position until manually changed.

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3.7 Drive spindle. The drive spindle shall have a 1/4 inch hexagon socket or a 1/4 inch square male drive tang as specified. Sockets and tangs shall conform to the requirements of ANSI B107.4. The hexagon socket shall have an externally knurled, quick release mechanism to expedite the use of commercial bits conforming to ANSI B107.4.

3.8 Clutch. The clutch shall be the internally adjustable type and shall be capable of controlling the torque output within the specified ranges and accuracies with the use of not more than one adjustment spring. The clutch shall completely disengage (i.e. impart no more torque to the driven fastener) when the preset torque is reached and shall automatically reengage when the tool is removed from the fastener being driven.

3.9 Air inlet connection. The air inlet connection shall consist of a well secured brass or ferrous fitting threaded with internal American standard taper pipe threads (NPT) or internal dryseal American standard taper pipe threads (NPTF). The air inlet shall be fitted with an air strainer or screen (20 mesh or finer) which shall effectively retain solid particles in the compressed air supply. The strainer or screen shall be of such design that it can be easily removed, cleaned, and replaced.

3.10 Carrying case. The nutrunner and screwdriver shall be furnished in a metal carrying case. The case shall be large enough to contain the tool and with a minimum of 10 cubic inches of space for the accessories. The case shall be made of 0.032 inch (minimum) thick steel with formed or welded corners, a heavy duty snap latch, and a suitable carrying handle of formed metal secured by welding, screws, or rivets. All surfaces of the case shall be painted and the case shall not open when carried by the handle.

3.11 Instruction manual. An instruction manual shall be furnished with each nutrunner and screwdriver. The manual shall contain operating and maintenance instructions, including instructions for lubrication and a spare parts list.

3.12 Identification marking. Each nutrunner and screwdriver shall be marked in a permanent and legible manner. The marking shall include the manufacturer's name or trademark and the model, size, or type designation so that the nutrunner and screwdriver and the source of manufacture may be readily identified.

3.13 Workmanship. All components of the nutrunner and screwdriver and the case shall be free of any defect that may affect the servicability, operation, or life expectancy. All paint shall be applied so as to produce a uniform surface free of runs, wrinkles, or blisters.

3.14 Type I. The type I nutrunner and screwdriver shall be similar to figure I and have a pistol grip handle. The drive spindle shall have a 1/4 inch hexagon socket or a 1/4 inch square drive tang as specified (See 6.2), conform to the applicable requirements specified in table I and be capable of completing the specified tests. The no load speed shall not be less than 750 RPM or exceed 1350 RPM with air consumption of 27 CFM maximum at maximum no load speed, weigh not more than 5 pounds, and not exceed 13 inches in overall length.

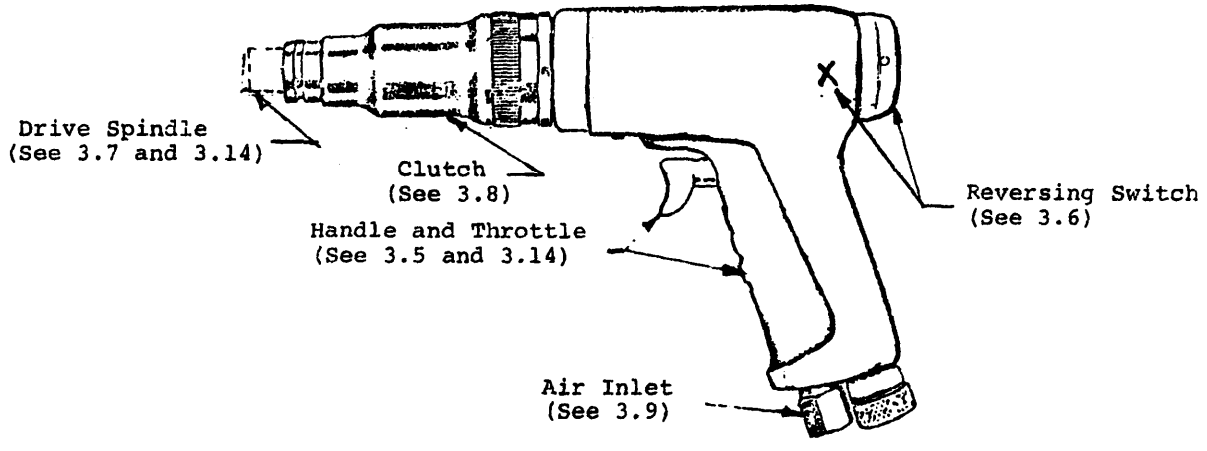


FIGURE I
Type I - Nutrunner and Screwdriver

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TABLE I

Size	Torque range (minimum)
1	10 to 50 inch-pounds.
2	25 to 90 inch-pounds.

3.15 Type II. The type II nutrunner and screwdriver shall be similar to figure II, have a body grip handle, a 1/4 inch hexagon socket drive spindle and develop a minimum torque range of 5 to 25 inch pounds and be capable of completing the specified tests. The no load spindle speed shall not be less than 500 RPM or exceed 1000 RPM with air consumption not exceeding 27 CFM at maximum no load speed, weigh not more than 4 pounds and not exceed 13 inches in overall length.

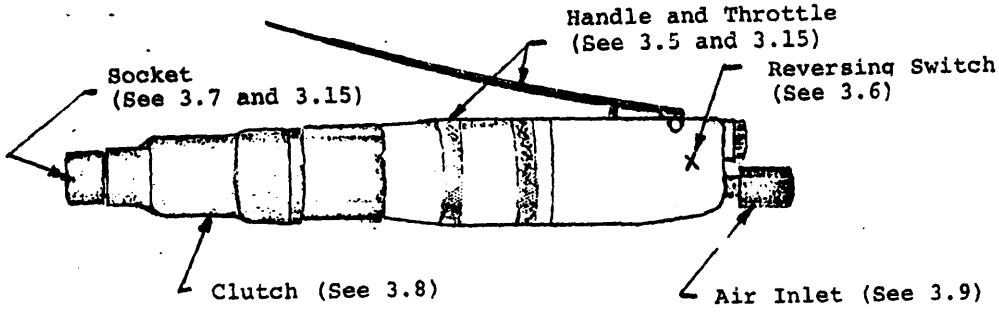


FIGURE II
Type II - Nutrunner and Screwdriver

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections:

(a) Preproduction inspection.
(b) Acceptance inspections.
(c) Preparation for delivery inspections.

4.3 Preproduction inspection. A preproduction sample of each type and size of nutrunner and screwdriver shall be subjected to all the examinations and tests specified herein. Presence of one or more defects shall be cause for rejection.

4.4 Acceptance inspection. Each sample nutrunner and screwdriver selected in accordance with 4.4.2 shall be examined as specified in 4.6 and tested as specified in 4.7.2 and 4.7.3 but excluding the disassembly specified in 4.7.3.1.

4.4.1 Lot. A lot for inspection purposes shall consist of all nutrunners and screwdrivers of the same type and size submitted for acceptance at the same time and place.

4.4.2 Sampling. Sampling for acceptance inspection shall be conducted in accordance with Inspection Level S-3 of MIL-STD-105 with an Acceptable Quality Level (AQL) of 2.5 percent defective.

4.5 Examination of preparation for delivery. Examination of preparation for delivery shall be in accordance with the quality assurance provisions of PPP-P-40.

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4.6 Examination. Each nutrunner and screwdriver shall be examined for conformance to all the requirements specified herein except those requirements covered by a specific test in 4.7 thru 4.7.6. The case and instruction manual shall be examined for conformance to the requirements specified in 3.10 and 3.11.

4.7 Tests. Any instrumentation used to test the speed, air consumption, torque, or sound shall have a current and valid certification as to its accuracy. Accuracy must be commensurate with tests being performed. The sequence of tests shall be as follows: performance, accuracy, service, sound, air leakage, and accuracy. Any tool that fails any test shall be considered defective.

4.7.1 Test conditions. All tests shall be conducted with the tool connected by a 10 foot length of 5/8 inch inside diameter hose to an air supply of 90 PSIG ± 2 PSIG measured as close to the tool as practicable. The air supply line shall be equipped with a filter and an air regulator and an in-line lubricator if recommended by the supplier.

4.7.2 Performance test. Each nutrunner and screwdriver shall be tested to determine conformance with the requirements specified in 3.5 and 3.6 for the throttle and reversing switch and the applicable speed and air consumption specified in 3.14 and 3.15.

4.7.3 Accuracy test. Each sample nutrunner and screwdriver shall be tested on a dynamic torque analyzer for repetitive torque setting accuracy at the applicable torque values specified in table II. Accuracy shall be tested with the tool operating in the clockwise (right hand) direction only. Fifty readings shall be made at each torque setting and the maximum spread in torque readings shall not exceed the values specified in table II.

TABLE II

Classification	Torque settings (inch-pounds ± 10%)	Maximum spread (inch-pounds)
Type I, Size 1	10	4
	30	6
	50	8
Type I, Size 2	25	5
	60	8
	100	10
Type II	5	2
	15	4
	25	5

4.7.3.1 When the accuracy test has been completed for the second time (see 4.7), the tool shall be disassembled and each component examined. Any evidence of excessive wear, broken, cracked, or otherwise deformed components shall be cause for rejection.

4.7.4 Service test. Each sample nutrunner and screwdriver shall be subjected to the following service test. The tool shall be operated manually or installed in a fixture in a manner that the tool can be loaded and automatically actuated for 1000 cycles. Each cycle shall be a minimum of 5 seconds on and a maximum of 5 seconds off. The load applied shall be the high limit of the specified torque range for the applicable tool. Any tool that fails to complete the 1000 cycles or the examination specified in 4.7.3.1 will be rejected.

4.7.5 Sound pressure level test. The sound pressure level test shall be conducted in accordance with the applicable provisions of ANSI S5.1, except as modified herein. The sound pressure level shall not exceed 90 decibels on the A weighted scale at any microphone location measured at an unobstructed distance of one meter.

4.7.5.1 Acoustical calibration. The entire instrumentation system shall be calibrated before each test series allowing a minimum instrument warm-up period of 5 minutes if transistorized and 1/2 hour if vacuum tube. Recalibration shall be performed if any instrument adjustment is necessary during progress of the test. All instrumentation (except microphone) and test personnel shall be as remotely located as practicable so that the measured data will be unaffected by a nonstationary observer.

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4.7.5.2 Background sound level. The ambient noise level shall be no greater than 10db less than the measurement taken with the tool running in all tool and microphone orientations. In addition, the sound produced by the energy absorbing device, used in the "on load" test, must be at least 10 dbA below the machine's own sound output.

4.7.6 Leakage test. Each sample nutrunner and screwdriver shall be tested for air leaking at or around the throttle in the closed position. The handle of the tool shall be submerged in water so that the throttle is below the water level. Any evidence of air bubbles emitting from the throttle will be cause for rejecting the tool.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The preservation, packaging, and packing shall be in accordance with PPP-P-40 at the level specified (See 6.2) and the marking shall be in accordance with PPP-P-40 as specified (see 6.2).

6. NOTES

6.1 Intended use. The nutrunner and screwdrivers covered herein are intended to be used to install threaded fasteners where repetitive torque setting accuracy is required. They are recommended for general maintenance and repair duty.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and size required (See 1.2.1).
(c) Specify if preproduction model is required (See 3.1).
(d) If type I is required, specify shape of drive spindle (See 3.14).
(e) Level of preservation, packaging and packing (see 5.1).
(f) Marking required (see 5.1).

6.3 The tools covered by this specification are in compliance with the requirements of the Occupational Safety and Health Administration (OSHA) as published prior to the date of this document.

6.4 Supersession data. This specification supersedes Military Specification MIL-N-26511C, dated 29 June 1973.

6.4.1 Cross reference data. The tools covered by this specification replace the four tools assigned National Item Identification Numbers which referenced the Military Specification MIL-N-26511, and the cross reference is as follows:

MIL-N-26511B	GGG-N-1907	MIL-N-26511C
Type I, Size 2	Type I, Size 1	None
Type I, Size 3	Type I, Size 2	None
Type II, Size 1	Type II	None
Type I, Size 3 (Hex socket)	Type I, Size 1 (Hex socket	26511/3
None	None	26511/2
None	None	26511/1

MILITARY INTEREST:

Custodians:

Army - GL(MCA)
Navy - SH
Air Force - 84

User activities:

Army - WC, AV, CE
Navy - AS, YD

Preparing activity:

GSA-FSS

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Price 30 cents each.

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